MATH 122 — Calculus II

Exam 2 Practice 4: Techniques of Integration

Name: _____ Date: _____

Instructions:

- **Show all work:** Full credit requires complete, correct steps and proper math notation.
- **Points:** Each question is worth 10 points.
- **Formula Sheet:** Permitted (formulas only; no worked examples).
- **Calculators:** Basic, non-graphing calculators allowed.

Part I: Integration by Parts

1. (10 points) Evaluate

$$I = \int x^6 \ln x \, dx$$

2. (10 points) Evaluate

$$I = \int x^3 \cos 2x \, dx$$

3. (10 points) Evaluate

$$I = \int \cot^{-1} x \, dx$$

4. (10 points) Evaluate the definite integral

$$\int_1^e \sqrt{x} \ln x \, dx$$

Part II: Trigonometric Integrals

5. (10 points) Evaluate

$$\int \cos^5 x \, dx$$

6. (10 points) Evaluate

$$I = \int \sin^3 x \cos^4 x \, dx$$

7. (10 points) Evaluate

$$I = \int \cos 4x \cos x \, dx$$

8. (10 points) Evaluate

$$\int 4 \sin x \sin 2x \cos 3x \, dx$$

Part III: Partial Fractions

9. (10 points) Evaluate

$$\int \frac{6}{(x-1)(2x+1)}\,dx$$

10. (10 points) Evaluate

$$\int \frac{4}{x^2(x+1)}\,dx$$

MATH 122 — Calculus II

Exam 2 Practice 5: Techniques of Integration

Name: _____ Date: _____

Instructions:

- **Show all work:** Full credit requires complete, correct steps and proper math notation.
- **Points:** Each question is worth 10 points.
- **Formula Sheet:** Permitted (formulas only; no worked examples).
- **Calculators:** Basic, non-graphing calculators allowed.

Part I: Integration by Parts

1. (10 points) Evaluate

$$I = \int \frac{\ln x}{\sqrt{x}}\, dx$$

2. (10 points) Evaluate

$$I = \int x^3 e^{-x}\, dx$$

3. (10 points) Evaluate

$$I = \int \tan^{-1}(2x)\, dx$$

4. (10 points) Evaluate the definite integral

$$\int_0^{\pi/2} x \cos x\, dx$$

Part II: Trigonometric Integrals

5. (10 points) Evaluate

$$\int \cos^6 x\, dx$$

6. (10 points) Evaluate

$$I = \int \sin^2 x \cos^3 x \, dx$$

7. (10 points) Evaluate

$$I = \int \sin 4x \cos 3x \, dx$$

8. (10 points) Evaluate

$$\int 4 \sin x \sin 2x \sin 3x \, dx$$

Part III: Partial Fractions

9. (10 points) Evaluate

$$\int \frac{10}{(x+2)(2x-1)}\,dx$$

10. (10 points) Evaluate

$$\int \frac{2}{x(x-1)^2}\,dx$$